

Do Language Models Honor Their Own Definitions? Measuring Definition Application Consistency

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Abstract

Large language models (LLMs) are increasingly deployed in multi-turn settings where a definition introduced in one turn should remain binding in subsequent reasoning. Little is known, however, about whether models actually respect such self-generated constraints. We introduce *definition–application consistency*: an evaluation framework that measures whether a model applies a definition it has previously generated when answering a subsequent, logically entailed query. Using a manually verified dataset of 1,000 pairs across six semantic domains (complemented by a 10,000-pair automatically generated set for scale analysis), we evaluate four contemporary LLMs spanning proprietary and open-weight families: GPT-4-Turbo, Claude 3 Opus, LLaMA-3-70B-Instruct, and Mistral-7B-Instruct (snapshot of October 2024). Large models are largely consistent on average (92–94%), but the residual 6–8% of failures is not uniformly distributed—it concentrates in procedural, scientific, and social-ethical concepts, and grows substantially under adversarial definitions and longer conversational contexts. A definition-grounded prompting intervention intended to enforce semantic commitments counterintuitively reduces consistency and increases hedging across models. This negative result suggests that semantic commitments induced through natural language are not reliably treated as binding constraints.

1 Introduction

In multi-turn deployments of large language models (LLMs) (Brown et al., 2020; OpenAI, 2023; Touvron et al., 2023), a model often introduces or explains a concept and is later asked to apply it. This pattern is everywhere in real deployments, and it carries concrete risks when it breaks. A medical assistant that defines a diagnostic criterion but inconsistently applies it across follow-up questions may steer clinicians toward unsafe decisions. A policy assistant that articulates an eligibility rule

but fails to enforce it in subsequent case checks can produce unfair recommendations even when each individual response sounds fluent. A tutoring system that teaches a concept and then contradicts it in the next turn gives students an incoherent signal. The common structure across these cases is simple: a constraint is introduced through natural language, and it must remain binding in later reasoning. Whether models actually do this is a prerequisite for trusting them in any multi-turn deployment, and it is what we set out to measure.

Why this is non-trivial. Model-generated definitions function, conversationally, as temporary semantic commitments, and a coherent agent should apply them consistently in subsequent reasoning. This is not the standard NLI setup. In benchmarks like SNLI (Bowman et al., 2015) and MultiNLI (Williams et al., 2018), premises come from outside the model and are assumed to be ground truth. Here, the premise is generated by the model itself, and what we measure is whether it can stay consistent with constraints it introduced—not whether it can reason from facts handed to it. The distinction matters because globally learned priors can conflict with locally introduced definitions, and those conflicts are exactly what users encounter when they try to correct, customize, or specialize a model’s behavior at inference time.

Prior work on LLM consistency has mostly looked at robustness to paraphrasing, repeated questioning, or explicit contradiction detection (Elazar et al., 2021; Lin et al., 2022; Mündler et al., 2024; Wang et al., 2023)—not at the interaction pattern where a model introduces a constraint through its own definition and later violates it.

Research questions.

- **RQ1:** Do current LLMs preserve semantic commitments introduced through their own generated definitions when answering a sub-

sequent entailed query?

- **RQ2:** If failures occur, are they random, or do they cluster systematically by semantic domain, conversational distance, or definition properties?
- **RQ3:** Can a lightweight, training-free prompting intervention that explicitly grounds the application step in the model’s own definition reduce inconsistency?

Illustrative example. Consider a two-turn exchange in which a model defines a triangle as “a shape with four equal sides.” Although incorrect with respect to real-world geometry, this definition induces a temporary semantic commitment. If asked “Is a square a triangle?” the logically consistent answer relative to the model’s definition is *yes*. Models, however, often answer *no*, revealing a violation of the constraint they themselves introduced.

Contributions.

- We formalize *definition–application consistency* as a diagnostic evaluation of whether language models preserve semantic commitments introduced through their own generated definitions.
- We construct a dataset of 1,000 definition–application pairs spanning six semantic domains with strict entailment verification, complemented by a larger automatically generated dataset for robustness analysis.
- We evaluate four proprietary and open-weight models and identify four systematic failure modes: definition forgetting, scope drift, over-generalization, and prior-knowledge override.
- We show that definition-grounded prompting interventions can counterintuitively degrade semantic consistency—a negative but informative result with implications for the broader reasoning-prompting literature.

2 Related Work

2.1 Consistency and Self-Contradiction

Prior work has examined consistency in language models by probing response stability under paraphrased, repeated, or reordered queries (Elazar et al., 2021; Lin et al., 2022; Wang et al., 2023;

Webson and Pavlick, 2022). These studies show that models often produce divergent answers despite semantic equivalence, revealing sensitivity to prompt phrasing and context ordering. Such evaluations primarily measure answer invariance rather than adherence to semantic constraints introduced within a conversation. Mündler et al. (2024) target self-contradiction by detecting mutually inconsistent factual claims within a single output; while effective for surface-level contradictions, these methods do not capture cases in which a model introduces a semantic constraint such as a definition and later violates it without producing an explicit contradiction. Our work targets this overlooked interaction pattern.

2.2 Faithfulness of Explanations

A parallel line of research studies the faithfulness of model-generated explanations (Jacovi and Goldberg, 2020; Rajani et al., 2019; Wiegrefe and Marasović, 2021), showing that explanations can be plausible yet disconnected from the reasoning that produces the final answer. Turpin et al. (2023) demonstrate that chain-of-thought rationales can be systematically biased by features the model fails to mention; Lanham et al. (2023) show that explanation length and content do not always reflect what drives the model’s prediction. Our setting differs in two key respects. First, the definitions are elicited *before* the application query and thus function as forward-looking semantic commitments rather than post-hoc justifications. Second, we evaluate behavioral consistency with respect to the content of the definition, rather than the faithfulness of explanation to internal computation.

2.3 Semantic Entailment and Dialogue Commitments

NLI benchmarks such as SNLI (Bowman et al., 2015) and MultiNLI (Williams et al., 2018) evaluate entailment relations between curated premises and hypotheses assumed to be ground truth. Our premises, in contrast, are generated dynamically by the model itself and treated as temporary semantic commitments; consistency is evaluated relative to these self-generated constraints. Work on multi-step compositional reasoning (Press et al., 2023) and long-context information use (Liu et al., 2024) further shows that models do not always carry information forward reliably. Holistic suites such as HELM (Liang et al., 2023) cover a wide range of single-turn capabilities but do not target preserva-

tion of commitments across turns.

3 Task Definition

We define a **definition–application pair** as a two-turn interaction:

1. **Definition Prompt:** the model is prompted to define a concept.
2. **Application Prompt:** the model is asked a binary question whose correct answer is logically entailed by the generated definition.

Let D denote the definition produced by the model and A its subsequent application answer. A pair is **consistent** if A is logically entailed by the semantic constraints expressed in D , and **inconsistent** otherwise. Note that we do *not* require D to be objectively correct with respect to external world knowledge; D is a temporary semantic commitment, and consistency is judged relative to that commitment, not to any external ground truth. The judgments are binary, but the task isolates a necessary condition for semantic reliability in interactive language use—any coherent multi-turn interaction requires that constraints introduced earlier be respected in subsequent reasoning. Failures at this level can propagate to more complex reasoning behaviors regardless of how fluent the generation looks.

Two evaluation regimes. Our benchmark contains two complementary regimes that share the same consistency criterion but differ in the relationship between the elicited definition and external world knowledge:

- **Standard regime** (main benchmark, 1,000 pairs). The model is asked to define a concept naturally. The resulting definitions are typically correct, and the application query tests whether the model applies its own (usually accurate) definition coherently.
- **Adversarial regime** (Appendix J). The model is given a definition that intentionally conflicts with widely held world knowledge (e.g., “A triangle is a shape with four equal sides”). The logically consistent answer follows from the stated definition but contradicts the global prior. This regime isolates whether models override locally introduced commitments with globally learned priors.

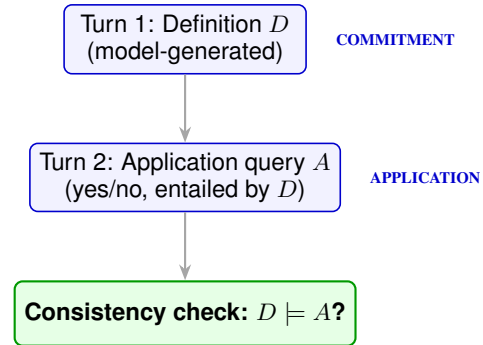


Figure 1: Definition–application consistency framework. A model-generated definition D (Turn 1) induces a semantic commitment that must logically entail ($\models$) the subsequent application A (Turn 2).

In both regimes, a response is consistent if and only if A is entailed by D . Unless otherwise noted, headline numbers refer to the standard regime; adversarial results are reported separately.

4 Dataset Construction

Domains and concept sourcing. The benchmark contains 1,000 manually verified definition–application pairs spanning six semantic domains: *common nouns* (WordNet high-frequency concrete entities), *scientific concepts* (introductory physics, chemistry, and biology from Wikipedia category pages and OpenStax indices), *mathematical/logical concepts* (undergraduate-level geometry, number theory, propositional logic), *procedural concepts* (algorithmic terms from CS textbook indices), *social/ethical concepts* (introductory ethics, sociology, political science glossaries), and *abstract properties* (e.g., *symmetric*, *transitive*, *reversible*). Concept lists were de-duplicated, lower-cased, and filtered to remove proper nouns and ambiguous polysemes. Per-domain counts and additional examples are provided in Appendix F and Appendix E.

Construction protocol. For each concept, a language model was prompted with Provide a concise definition of the concept [X] to generate D . We then constructed an application query asking whether a specific instance satisfies the semantic constraints expressed in D , requiring a binary (yes/no) judgment. Each pair was manually verified by two trained annotators with graduate-level NLP training, with disagreements adjudicated by a third annotator (inter-annotator agreement Cohen’s $\kappa = 0.84$ on entailment direction). Verification ensured the correct answer was logically entailed by D , not by external world knowledge. Definitions range from concise taxo-

Definition generated)	(model-	Application Query	Consistent	Inconsistent
A mammal is an animal that has fur, is warm-blooded, and feeds milk to its young.		Is a dog a mammal?	Yes	No
An algorithm is a finite sequence of instructions that terminates.		Is an infinite loop an algorithm?	No	Yes
A triangle is a shape with four equal sides.		Is a square a triangle?	Yes	No

Table 1: Example pairs from the benchmark. The “Consistent” column shows the answer entailed by the stated definition. The third row illustrates the adversarial regime.

nomic statements to relational and normative descriptions; application queries include both direct instantiation checks and boundary cases that stress semantic constraints. The dataset pipeline is illustrated in Appendix B (Figure 2).

Consistency labeling. We adopt a *hybrid* labeling protocol to avoid reliance on an LLM judge (which would re-introduce the very capability we are evaluating). (i) A rule-based extractor parses each application response into {*yes*, *no*, *unclear*} via case-insensitive prefix match on the first sentence with a hedged-expression fallback. (ii) The extracted answer is compared to the entailed answer recorded during dataset construction; *unclear* is treated as inconsistent in the binary metric but separated in the three-way analysis. (iii) A random 20% sample, plus all *unclear* responses, were re-labeled by two graduate annotators (Cohen’s $\kappa = 0.91$ on consistency labels), confirming alignment with human judgments.

Verification trade-off. Each instance is manually verified, prioritizing semantic precision over automated scale. Observed inconsistencies thus cannot be attributed to annotation noise or ambiguous entailment, and represent a conservative lower bound on definition–application failures.

5 Models and Experimental Setup

We evaluate four representative language models spanning proprietary and open-weight families, to probe whether definition–application inconsistencies arise across different model scales, training regimes, and access constraints.

Proprietary models. **GPT-4-Turbo** (OpenAI, 2023) (gpt-4-turbo-2024-04-09, OpenAI API, October 2024) and **Claude 3 Opus** (Anthropic, 2024) (claude-3-opus-20240229, Anthropic API, October 2024).

Model	Entailed	Contradicted	Unclear
GPT-4-Turbo	93.5	6.0	0.5
Claude 3 Opus	93.8	6.2	0.0
LLaMA-3-70B-Instruct	94.3	5.7	0.0
LLaMA-3-70B-Instruct (DGR)	84.9	8.3	6.8
Mistral-7B-Instruct	50.2	49.8	0.0

Table 2: Three-way breakdown of application responses. Definition-grounded reasoning (DGR) increases hedging (unclear responses) without improving semantic consistency.

Open-weight models. **LLaMA-3-70B-Instruct** (Meta) and **Mistral-7B-Instruct-v0.2** (Jiang et al., 2023), downloaded from Hugging Face in October 2024. These models are deployed using standard inference settings without task-specific fine-tuning. Because no peer-reviewed publication describes LLaMA-3-70B-Instruct, we cite the closest available reference (Touvron et al., 2023) and rely on the official model release notes for version details.

Evaluation protocol. All models are queried using identical definition and application prompts. We use greedy decoding (temperature 0.0, top- p 1.0, max 128 tokens) to minimize sampling variability. Application queries require binary (yes/no) responses to enable direct comparison across models and settings.

6 Evaluation Metrics

Our primary metric is **definition–application consistency**: the proportion of instances in which the model’s application answer is logically entailed by its own generated definition, judged using the hybrid protocol of §4. Ambiguous or hedged responses are treated conservatively as inconsistent. We additionally analyze a three-way categorization of responses into *entailed*, *contradicted*, and *unclear* to disentangle explicit semantic violations from uncertainty-driven failures (Table 2).

7 Results

Large models achieve high overall consistency in the standard regime (92–94%), but structured failures persist in specific domains and prompting conditions. Scientific and procedural concepts show the highest inconsistency rates across all large models (Table 3), which suggests that applying multi-constraint definitions remains the hardest part of the task. We analyze these patterns qualitatively in §9.

Effect of model scale. To assess whether consistency depends on model capacity and persists

Domain	GPT-4-Turbo	Claude 3	LLaMA-3	Mistral-7B
Common Nouns	95.7	95.4	96.1	55.0
Scientific Concepts	90.9	90.7	91.2	48.3
Mathematical / Logical	94.1	93.9	94.8	51.6
Procedural Concepts	90.4	90.1	89.5	47.2
Social / Ethical	92.6	92.1	92.4	49.8
Abstract Properties	94.9	94.6	95.3	50.9

Table 3: Definition–application consistency (%) across semantic domains.

Setting	Consistency
With Definition	94.3%
Definition-Free	68.0%
2-turn (baseline)	94.3%
3-turn (+1 neutral turn)	89.1%
4-turn (+2 neutral turns)	86.7%

Table 4: Top: removing access to the model’s own definition (LLaMA-3-70B-Instruct). Bottom: effect of conversational distance, inserting neutral intervening turns between the definition and the application query. Both manipulations substantially reduce consistency.

beyond manually curated data, we evaluate Mistral-7B-Instruct on the 10,000-pair automatically generated dataset (Appendix H). Despite producing fluent outputs, Mistral achieves only 50.2% consistency—approximately chance on a binary entailment task.

Definition-free baseline. To verify that consistency arises from genuine definition grounding rather than task simplicity or prior world knowledge, we evaluate a *definition-free* setting in which the model sees only the application query. For LLaMA-3-70B-Instruct, consistency drops from 94.3% to 68.0% (Table 4), confirming that the task requires preserving the locally induced commitment rather than defaulting to global priors.

Conversational distance. We additionally test whether semantic commitments degrade as conversational context grows by inserting neutral turns—unrelated to the concept—between the definition and the application query. Consistency drops from 94.3% in the baseline 2-turn setting to 86.7% at 4 turns (Table 4, bottom), indicating that commitments become increasingly unstable with conversational distance, related to but distinct from prior observations on long-context degradation (Liu et al., 2024).

Bootstrap confidence intervals, adversarial-definition results, and prompt-robustness checks are reported in Appendices I, J, and K.

Human baseline. To contextualize model performance, three graduate-level annotators judged a random 200-pair subset using only the model-

Evaluator	Consistency (%)
Human (3 annotators)	99.3
GPT-4-Turbo	93.5
Claude 3 Opus	93.8
LLaMA-3-70B-Instruct	94.3
Mistral-7B-Instruct	50.2

Table 5: Human vs. model consistency under identical information constraints.

generated definition. Humans achieved 99.3% consistency (Table 5), confirming that model errors reflect failures in semantic-commitment preservation rather than task ambiguity.

8 Mitigation: A Negative Result

We test a widely assumed hypothesis: explicitly grounding reasoning in the model’s own definition should improve semantic consistency. The hypothesis sits naturally alongside the broader literature on reasoning-style prompting—chain-of-thought (Wei et al., 2022; Kojima et al., 2022), self-consistency (Wang et al., 2023), least-to-most (Zhou et al., 2023), and self-refine (Madaan et al., 2023).

Definition-grounded reasoning (DGR). The application prompt explicitly instructs the model to rely solely on its previously generated definition when answering the query. This intervention is intended to foreground local semantic commitments and suppress interference from global world knowledge. We evaluated several prompting variants and report results for DGR, the most stable across models.

Mitigation results. DGR *reduces* consistency, contrary to expectations. For LLaMA-3-70B-Instruct, accuracy drops from 94.3% to 84.9%, while the rate of *unclear* responses rises to 6.8% (Table 2). The degradation is most pronounced for scientific, procedural, and social concepts, and most severe in smaller models, where explicit grounding does not compensate for weaker implicit representations. This echoes prior findings that chain-of-thought rationales do not always faithfully reflect the underlying decision process (Lanham et al., 2023; Turpin et al., 2023), and gives a concrete case in which a reasoning-style intervention systematically harms rather than helps task performance.

We frame this as a useful negative result. A commonly suggested intervention—explicitly grounding reasoning in the model’s own definitions—can

Domain	Forget	Drift	Over-gen	Override
Common Nouns	12%	6%	5%	2%
Procedural	31%	22%	14%	9%
Scientific	28%	25%	11%	12%
Social/Ethical	14%	10%	8%	24%

Table 6: Distribution of error mechanisms by domain. Percentages are the proportion of inconsistent cases attributable to each failure mode. *Forget*: definition forgetting; *Drift*: scope drift; *Over-gen*: overgeneralization; *Override*: implicit assumption override.

Failure Type	Concrete Example
Implicit Assumption Override	Definition: “A triangle is a shape with four equal sides.” Application: “Is a square a triangle?” Model Answer: No (global prior overrides local definition).
Definition Forgetting	Definition: “A bird is a vertebrate animal with feathers and wings.” Application: “Is a penguin a bird?” Model Answer: No (forgets stated criteria).

Table 7: Representative qualitative examples of definition–application inconsistency. In both cases, the model’s application answer contradicts its own generated definition.

systematically degrade semantic consistency. Identifying the limits of such interventions matters because it points future work toward mechanisms that genuinely preserve conversational commitments rather than merely instructing models to do so.

Definition quality and length. Inspecting 100 randomly sampled definitions, over 91% were judged well-formed, indicating that most failures arise during the application step rather than from defective definitions. Incorrect application answers are also rarely accompanied by explicit uncertainty markers. Analysis by definition length reveals a non-monotonic pattern: medium-length definitions yield the highest consistency, while both shorter and longer definitions perform worse, suggesting a trade-off between underspecification and cognitive overload.

9 Error Analysis

Qualitative analysis of inconsistent cases reveals structured and recurring failure mechanisms rather than random errors. We categorize failures into four dominant types, with their distribution across domains shown in Table 6 and illustrative examples in Table 7.

Definition forgetting occurs when the model

states correct definitional criteria but fails to apply them at the application step (e.g., defining birds as feathered vertebrates yet denying that a penguin is one). *Scope drift* involves subtle alteration of the stated constraints (e.g., classifying a non-terminating procedure as an algorithm after defining algorithms as finite). *Overgeneralization* extends a definition too broadly (e.g., labeling chemically neutral substances as acids after defining an acid as a proton donor). *Implicit assumption override* occurs when responses are dominated by prior world knowledge rather than the locally stated definition; this is especially prevalent for social and ethical concepts. Because definitions are generated immediately before the application query, these failures are not driven by long-term memory decay but by instability in local semantic reasoning.

10 Discussion

Interpreting the headline numbers. The headline numbers admit two readings. Optimistically, large models are mostly consistent: 92–94% in the standard regime, well above chance and not far below the 99.3% human ceiling. The cautionary reading is that the residual 6–8% is not noise. It clusters in procedural and scientific concepts (where definitions impose multiple constraints), in social and ethical concepts (where prior beliefs override the stated definition), under longer conversational contexts (down to 86.7% at distance four), and under adversarial definitions, where models routinely revert to their global priors. So failure is structured, and it concentrates in the cases that matter most for high-stakes deployment. When we describe these as *systematic violations*, we don’t mean models fail most of the time—we mean that when they do fail, they fail predictably and in the most consequential places.

Why definition-grounded prompting backfires.

The DGR result is the most striking finding in the paper, and it deserves discussion. We designed the intervention to be the obvious mitigation for the failure mode the benchmark measures: if a model fails to apply its own definition, telling it to do so should help. Instead, accuracy falls and hedging rises. One reading—consistent with Lanham et al. (2023) and Turpin et al. (2023)—is that the implicit mechanisms by which models apply locally introduced constraints are not the same as those engaged by explicit step-by-step reasoning. Forcing the model into a reasoning frame may surface tensions

between the stated definition and the model’s global priors that would otherwise have been suppressed, producing either confident overrides (lower accuracy) or visible hedging (more *unclear* responses). If this is right, then reliable commitment preservation likely requires architectural or training-time interventions, not prompt-level instructions to “be consistent.”

Beyond binary entailments. The evaluation uses binary judgments, but the setting isolates a minimal and necessary requirement for coherent interaction: once a model introduces a semantic constraint, subsequent reasoning should respect it. Many real-world failures—inconsistent explanations, shifting definitions, policy misapplication—decompose into violations of this basic property. Taken together, the results indicate that definition–application inconsistency is not an artifact of dataset size or task design but a fundamental interaction-level limitation of current language models.

11 Conclusion

We introduced definition–application consistency as a diagnostic evaluation of whether language models reliably apply semantic constraints they introduce themselves. Across four model families, high single-turn accuracy coexists with structured failures to honor model-generated definitions, even in minimal two-turn interactions. A prompting intervention intended to enforce semantic grounding degrades performance instead. Together, these findings indicate that commitment preservation is not guaranteed by fluent generation or by explicit reasoning prompts, and they motivate interaction-aware evaluation frameworks and commitment-preserving mechanisms in future language models.

Limitations

Dataset size. Our manually verified benchmark consists of 1,000 pairs. While this prioritizes semantic precision over scale, larger curated datasets could enable finer-grained linguistic analyses. We complement this with an automatically generated 10,000-pair dataset used only for scale sensitivity analysis (Appendix H).

Model snapshot. Results for GPT-4-Turbo and Claude 3 Opus reflect API behavior as of October 2024. Because providers update these models without changing the public identifier, exact reproduc-

tion at later dates may not be possible. We also do not evaluate newer frontier models that have been released since the snapshot; whether the patterns reported here persist in later-generation models is an open empirical question and an obvious direction for follow-up work.

Training-time interventions. We focus on prompting strategies and do not explore fine-tuning (Devlin et al., 2019; Ouyang et al., 2022) or explicit memory mechanisms. Although DGR degrades performance, this does not rule out architectural or training-based approaches.

Interaction length. Our setting consists of minimal two-turn interactions (extended to four turns in the distance experiment). Longer conversations may introduce additional forms of semantic drift not captured here (Liu et al., 2024).

Task simplicity. We use binary (yes/no) judgments to isolate a minimal and necessary condition for semantic reliability. This abstraction does not capture the full nuance of open-ended reasoning or graded responses.

English-only evaluation. All experiments are conducted in English. Whether the observed failure modes transfer to other languages—particularly low-resource ones—remains open.

References

- Anthropic. 2024. The Claude 3 model family: Opus, Sonnet, Haiku. Technical report (model card), Anthropic. https://www-cdn.anthropic.com/de8ba9b01c9ab7cbabf5c33b80b7bbc618857627/Model_Card_Claude_3.pdf.
- Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. 2015. A large annotated corpus for learning natural language inference. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 632–642, Lisbon, Portugal. Association for Computational Linguistics.
- Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language models are few-shot learners. In

- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
- Yanai Elazar, Nora Kassner, Shauli Ravfogel, Abhilasha Ravichander, Eduard Hovy, Hinrich Schütze, and Yoav Goldberg. 2021. Measuring and improving consistency in pretrained language models. *Transactions of the Association for Computational Linguistics*, 9:1012–1031.
- Alon Jacovi and Yoav Goldberg. 2020. Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4198–4205, Online. Association for Computational Linguistics.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L  lio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. 2023. Mistral 7B. *arXiv preprint arXiv:2310.06825*.
- Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. In *Advances in Neural Information Processing Systems* 35 (NeurIPS 2022), pages 22199–22213.
- Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, Kamil   Luko  i  t  , Karina Nguyen, Newton Cheng, Nicholas Joseph, Nicholas Schiefer, Oliver Rausch, Robin Larson, Sam McCandlish, Sandipan Kundu, Saurav Kadavath, Shannon Yang, Thomas Henighan, Timothy Maxwell, Timothy Telleen-Lawton, Tristan Hume, Zac Hatfield-Dodds, Jared Kaplan, Jan Brauner, Samuel R. Bowman, and Ethan Perez. 2023. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*.
- Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christian Cosgrove, Christopher D. Manning, Christopher R  , Diana Acosta-Navas, Drew A. Hudson, Eric Zelikman, Esin Durmus, Faisal Ladhak, Frieda Rong, Hongyu Ren, Huaxiu Yao, Jue Wang, Keshav Santhanam, Laurel Orr, Lucia Zheng, Mert Yuksekgonul, Mirac Suzgun, Nathan Kim, Neel Guha, Niladri Chatterji, Omar Khattab, Peter Henderson, Qian Huang, Ryan Chi, Sang Michael Xie, Shibani Santurkar, Surya Ganguli, Tatsunori Hashimoto, Thomas Icard, Tianyi Zhang, Vishrav Chaudhary, William Wang, Xuechen Li, Yifan Mai, Yuhui Zhang, and Yuta Koreeda. 2023. Holistic evaluation of language models. *Transactions on Machine Learning Research*.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, Dublin, Ireland. Association for Computational Linguistics.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-Refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems* 36 (NeurIPS 2023).
- Niels M  ndler, Jingxuan He, Slobodan Jenko, and Martin Vechev. 2024. Self-contradictory hallucinations of large language models: Evaluation, detection and mitigation. In *The Twelfth International Conference on Learning Representations (ICLR)*.
- OpenAI. 2023. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems* 35 (NeurIPS 2022).
- Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. 2023. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711, Singapore. Association for Computational Linguistics.
- Nazneen Fatema Rajani, Bryan McCann, Caiming Xiong, and Richard Socher. 2019. Explain yourself! Leveraging language models for commonsense reasoning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*

(ACL), pages 4932–4942, Florence, Italy. Association for Computational Linguistics.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Ranjan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurelien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. 2023. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations (ICLR)*.

Albert Webson and Ellie Pavlick. 2022. Do prompt-based models really understand the meaning of their prompts? In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 2300–2344, Seattle, United States. Association for Computational Linguistics.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, pages 24824–24837.

Sarah Wiegrefe and Ana Marasović. 2021. Teach me to explain: A review of datasets for explainable NLP. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks 1 (NeurIPS Datasets and Benchmarks 2021)*.

Adina Williams, Nikita Nangia, and Samuel R. Bowman. 2018. A broad-coverage challenge corpus for

sentence understanding through inference. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 1112–1122, New Orleans, Louisiana. Association for Computational Linguistics.

Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc Le, and Ed H. Chi. 2023. Least-to-most prompting enables complex reasoning in large language models. In *The Eleventh International Conference on Learning Representations (ICLR)*.

A Per-Model Domain Consistency

Domain	Consistency (%)
Common Nouns	96.1
Scientific Concepts	90.3
Mathematical / Logical	94.2
Procedural Concepts	89.7
Social / Ethical Concepts	91.4
Abstract Properties	95.0

Table 8: Definition–application consistency across semantic domains for LLaMA-3-70B-Instruct.

B Dataset Construction Pipeline

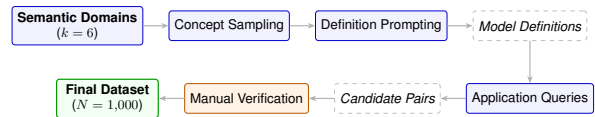


Figure 2: Workflow for the dataset construction pipeline.

C Implementation Details

Open-weight models were executed on NVIDIA A100 GPUs using the HuggingFace Transformers library with default inference configurations. Proprietary models were accessed through official APIs (OpenAI and Anthropic) in October 2024. All experiments used greedy decoding with temperature 0.0, top- p 1.0, and a maximum generation length of 128 tokens.

D Prompt Templates

Definition prompt:

Define the concept [X] in one sentence.

Application prompt:

Based on the definition above, is [instance] an example of [X]? Answer yes or no.

Definition-grounded reasoning prompt:

Use the definition above and reason step by step before answering the question.

E Additional Dataset Examples by Domain

Domain	Definition	Application Query
Common Nouns	A chair is a piece of furniture with a seat and a backrest for one person.	Is a sofa a chair?
Scientific	A catalyst is a substance that speeds up a chemical reaction without being consumed.	Is an enzyme a catalyst?
Math/Logical	A prime number is a natural number greater than 1 with exactly two divisors.	Is 7 a prime number?
Procedural	An algorithm is a finite sequence of instructions that terminates.	Is an infinite loop an algorithm?
Social/Ethical	A monopoly is a market in which a single supplier controls the entire supply.	Is a competitive market a monopoly?
Abstract	A relation is symmetric if whenever aRb holds, bRa also holds.	Is the "less than" relation symmetric?

Table 9: Representative definition–application pairs across all six semantic domains.

F Dataset Statistics

The benchmark contains 1,000 manually verified definition–application pairs. The average definition length is 14 tokens and the average application query length is 9 tokens. Lexical overlap between definitions and queries averages 0.21, indicating that models cannot rely on surface pattern matching.

Domain	# Examples
Common Nouns	170
Scientific Concepts	160
Mathematical / Logical	170
Procedural Concepts	170
Social / Ethical Concepts	160
Abstract Properties	170
Total	1,000

Table 10: Distribution of pairs across semantic domains.

G Annotation Procedure

Each definition–application pair was independently verified by two annotators with graduate-level training in NLP. Annotators determined whether the application query was logically entailed by the generated definition, relying only on the semantic constraints expressed in the definition. Disagreements

were resolved through adjudication with a third annotator. Inter-annotator agreement (Cohen’s κ) was 0.84 for entailment-direction labels and 0.91 for consistency labels on model responses (per §4). Ambiguous definitions were excluded during verification.

H Automatically Generated Dataset

To evaluate robustness beyond manually curated examples, we constructed an automatically generated dataset that is not manually verified and is used only for scale analysis (never for the headline numbers in the main text).

Construction pipeline. The automatic dataset is produced by a four-stage pipeline that mirrors the manual protocol but removes the human verification step:

- 1. Concept harvesting.** For each of the six semantic domains, we expand the seed concept list by sampling additional terms from the same source. De-duplication and the polyseme filter from the manual protocol are reapplied. This step yields approximately 12,000 candidate concepts.
- 2. Definition elicitation.** Each candidate concept is presented to the same definition prompt used for the manual dataset, with greedy decoding and a 128-token cap. Outputs that do not contain a full sentence or that contain the concept name verbatim more than twice are discarded.
- 3. Application-query generation.** For each surviving definition, an application query is generated by a templated procedure that pairs the concept with a candidate instance drawn from a hand-curated instance bank per domain. The entailed answer is computed automatically from a lightweight rule: the instance is positive if it satisfies all noun-phrase predicates extracted from the definition by a syntactic parser, and negative otherwise.
- 4. Pair filtering.** Pairs whose entailment label cannot be confidently determined by the rule, and pairs whose definition contradicts itself within a single sentence, are discarded. The final pool is sub-sampled uniformly per domain to produce 1k, 5k, and 10k splits.

Quality caveat. Because entailment labels are produced by a rule rather than by human annotation, the automatic dataset is noisier than the manually verified benchmark. We therefore use it only to study scale sensitivity, not to draw conclusions about absolute consistency. A spot-check of 200 randomly sampled auto-generated pairs found that 87% had unambiguous entailment labels, providing a rough upper bound on the headline numbers obtainable from this set.

H.1 Effect of dataset scale

We evaluate Mistral-7B-Instruct on subsets of 1k, 5k, and 10k pairs. Consistency improves only marginally with scale, rising from 0.51 at 1k to 0.58 at 10k. Despite a tenfold increase in data, performance remains far below large-model and human accuracy, consistent with prior reports of compositionality-related limitations (Press et al., 2023).

Auto Dataset Size	Consistency
1k	0.51
5k	0.54
10k	0.58

Table 11: Scale sensitivity for Mistral-7B-Instruct on the automatically generated dataset.

I Statistical Significance

We perform non-parametric bootstrap resampling with 10,000 iterations over definition–application pairs and report 95% confidence intervals.

Domain	Consistency	95% CI
Common Nouns	96.1	[94.0, 97.8]
Scientific Concepts	90.3	[87.2, 93.1]
Procedural Concepts	89.7	[86.5, 92.4]

Table 12: Bootstrap confidence intervals for domain-level consistency (LLaMA-3-70B-Instruct).

Differences across domains were considered statistically significant when bootstrap confidence intervals did not overlap.

J Adversarial Definition Test

To examine whether models prioritize globally learned knowledge over locally introduced semantic commitments, we construct adversarial definitions that intentionally conflict with real-world knowledge (e.g., “A triangle is a shape with four equal sides”). In these cases the logically consistent

answer contradicts real-world knowledge but follows from the definition. Across evaluated models, we observe frequent violations of these adversarial commitments: models often revert to prior semantic knowledge rather than applying the definition introduced earlier in the conversation, consistent with observations that rationales are sometimes overridden by prior beliefs (Turpin et al., 2023; Wiegrefe and Marasović, 2021).

K Prompt Robustness

We test multiple definition prompts (“Explain what the concept X means.”, “Provide a definition for X.”, “What is the meaning of X?”). Across prompt variants, consistency scores varied by less than two percentage points for large models, indicating that the observed failure patterns are not artifacts of a specific prompt formulation.

L Future Work

Future work could explore architectural mechanisms for tracking semantic commitments across conversational turns, such as explicit memory modules or symbolic constraint tracking. Evaluating later-generation frontier models, scaling the benchmark to longer interaction sequences, and extending to multilingual settings may further reveal the limits of commitment preservation in language models.